

Complex Numbers



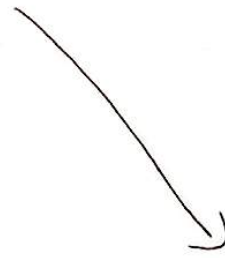
Reals

(e.g. -2 , $\sqrt[3]{5}$, $0.333\dots$, π).



Rationals

(e.g. -2 , $0.333\dots$, $\frac{7}{5}$)



Irrationals

(e.g. $\sqrt[3]{5}$, π , e)



Integers

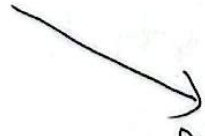
(e.g. -2 , 0 , 1 , 2)



Positive
Integers



0



Negative
Integers

Major Operations on Real Numbers:

$$+, \times, -, \div$$

Let's start with $+$, $\times$

These operations are both commutative
(i.e. $a+b=b+a$, $a \times b = b \times a$)

They are also associative

(i.e. $(a+b)+c = a+(b+c)$, $(a \times b) \times c = a \times (b \times c)$)

They both have identity elements, 0 and 1, respectively.

They both permit taking inverses, $a + (-a) = 0$
negatives and reciprocals, respectively.
 $a \cdot \frac{1}{a} = 1$

Multiplication distributes over addition
i.e. $a \times (b+c) = a \times b + a \times c$

Notation: $a \times b$ written as $a \cdot b$ or ab

For any real number a , $-a = (-1) \cdot a$

Question: Can we use these ideas to explain why $-(-6) = 6$?
And then why $(-2) \cdot (-3) = 6$?

Subtraction and division can be defined via addition and multiplication.

$$a - b = a + (-b)$$

$$a \div b = a \cdot \frac{1}{b}$$

 also written as $\frac{a}{b}$

a = "numerator"

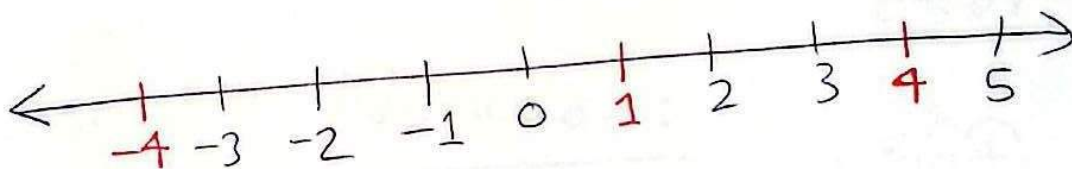
b = "denominator"

Question: How can we add fractions?

e.g. what is $\frac{2}{5} + \frac{(-1)}{3}$?

$$\frac{2}{5} - \frac{1}{3} = \frac{2 \cdot 3}{5 \cdot 3} - \frac{1 \cdot 5}{3 \cdot 5} = \frac{6 - 5}{15}$$

Order on Real Numbers: $= \frac{1}{15}$



Fill in Inequality Sign:

$$1 < 4$$

$$4 \geq 4$$

$$1 > -4$$

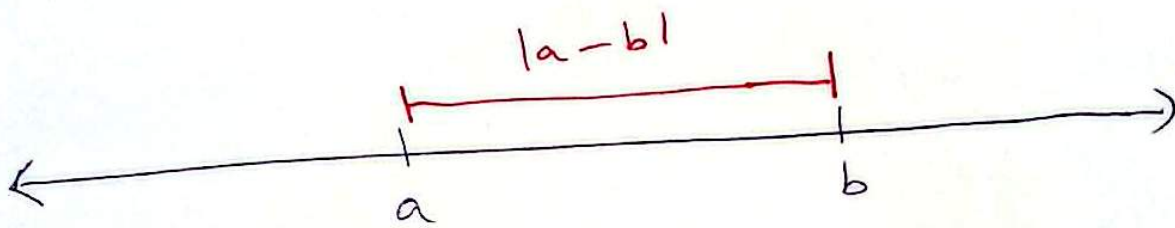
$$4 > -4$$

$$|4| \text{ and } |-4| = 4$$

Diagram showing the absolute value of 4 and -4. On the left, a circled 4 has an arrow pointing to -4. On the right, a circled -4 has an arrow pointing to 4.

("absolute value" of a) is the larger of a and $-a$

$|a-b|$ is a measure of distance between a and b



Scientific Notation:

$$1,004 = 1.004 \times 10^3$$

$$550 = \underline{5.50 \times 10^2}$$

$$-550 = \underline{-5.50 \times 10^2}$$

$$0.0305 = \underline{3.05 \times 10^{-2}}$$

$$\underline{0.0305}$$

$$\underline{5.50}$$

$$\underline{1,004}$$

$$\underline{550}$$

$$10$$

$$100$$

$$1000$$

$$10000$$

$$c \times 10^n$$

$$1 \leq c < 10$$